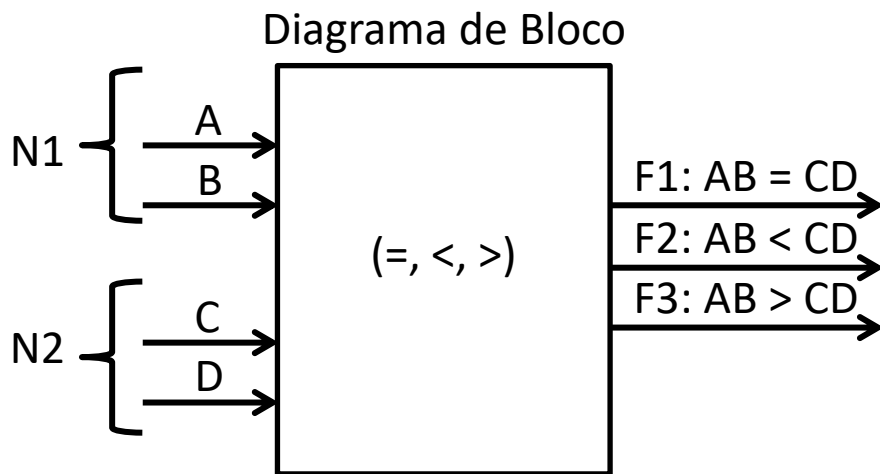


Atividade At-04 – Mapas de Karnaugh

- Coloque as funções F1, F2 e F3 na forma padrão de mintermos (soma de produtos).
- Represente as funções F1, F2 e F3 utilizando Mapas de Karnaugh.
- Simplifique as funções booleanas.
- Represente graficamente o circuito resultante.

L	A	B	C	D	F1	F2	F3	Mintermos
0	0	0	0	0	1	0	0	$A'B'C'D'$
1	0	0	0	1	0	1	0	$A'B'C'D$
2	0	0	1	0	0	1	0	$A'B'CD'$
3	0	0	1	1	0	1	0	$A'B'CD$
4	0	1	0	0	0	0	1	$A'BC'D'$
5	0	1	0	1	1	0	0	$A'BC'D$
6	0	1	1	0	0	1	0	$A'BCD'$
7	0	1	1	1	0	1	0	$A'BCD$
8	1	0	0	0	0	0	1	$AB'C'D'$
9	1	0	0	1	0	0	1	$AB'C'D$
10	1	0	1	0	1	0	0	$AB'CD'$
11	1	0	1	1	0	1	0	$AB'CD$
12	1	1	0	0	0	0	1	$ABC'D'$
13	1	1	0	1	0	0	1	$ABC'D$
14	1	1	1	0	0	0	1	$ABCD'$
15	1	1	1	1	1	0	0	$ABCD$



$$F1 = A'B'C'D' + A'BC'D + AB'CD' + ABCD$$

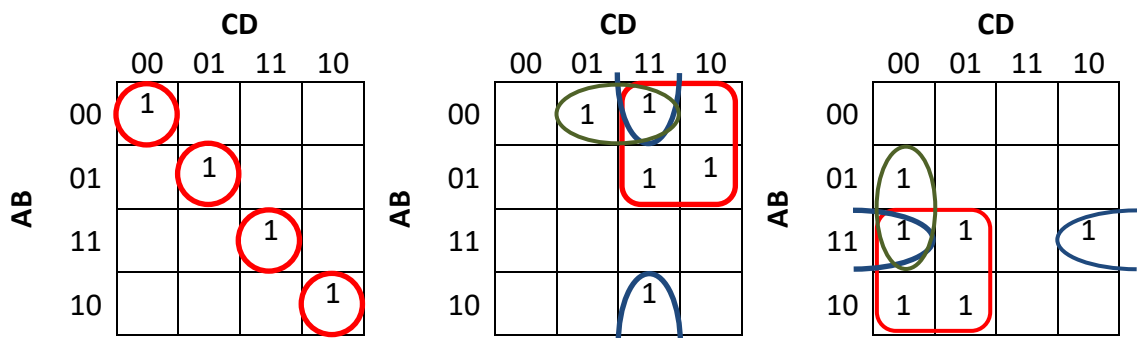
$$\text{Ou } F1 = \sum_{ABCD} (0, 5, 10, 15)$$

$$F2 = A'B'C'D + A'B'CD' + A'B'CD + A'BCD' + A'BCD + AB'CD$$

$$\text{Ou } F2 = \sum_{ABCD} (1, 2, 3, 6, 7, 11)$$

$$F3 = A'BC'D' + AB'C'D' + AB'C'D + ABC'D' + ABC'D + ABCD'$$

$$\text{Ou } F3 = \sum_{ABCD} (4, 8, 9, 12, 13, 14)$$



$$F1 = A'B'C'D' + A'BC'D + AB'CD' + ABCD$$

$$F2 = A'C + B'CD + A'B'D$$

$$F3 = AC' + BC'D' + ABD'$$

